החוג למערכות מידע
הפקולטה למדעי המחשב והמידע
אוניברסיטת חיפה

Software Engineering and Software Quality Assurance in Information Systems

Assignment 3

Prototype

For Ludo Educational Game

Panther Group

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Public Link: https://github.com/IslamHassadya11/SE-Ludo-Panther

Department of Information Systems

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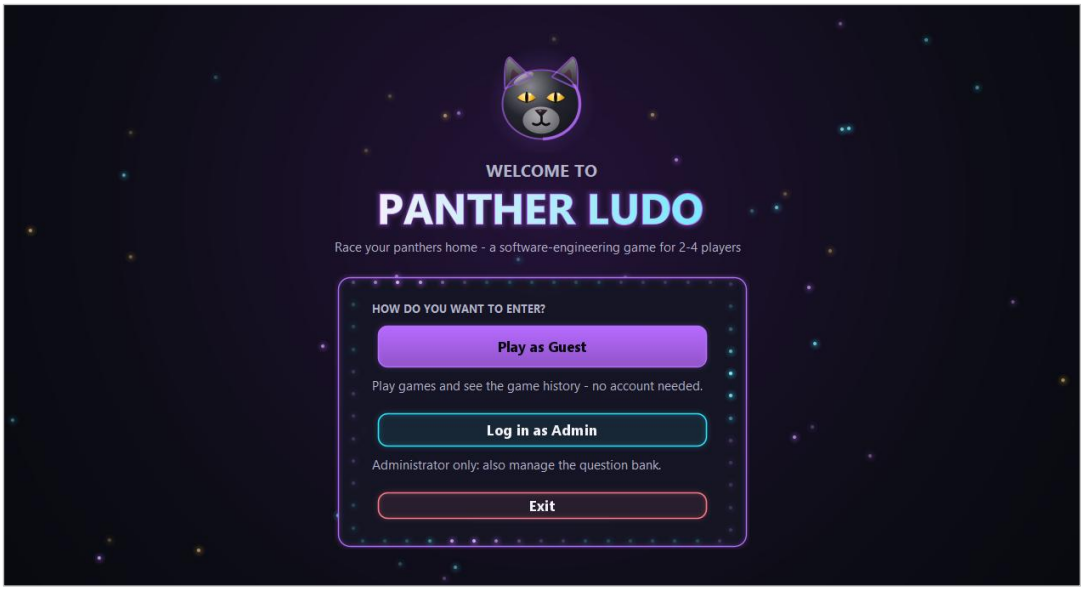
2026-2027

1. Process 1 – Playing a Game as a Guest

From opening the program to the final ranking.

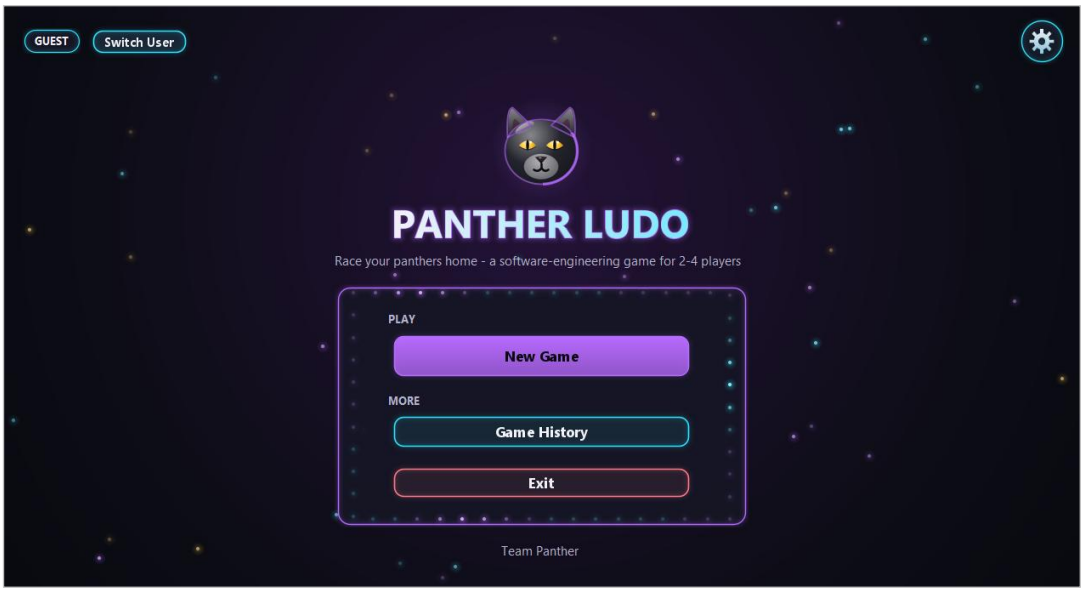
1.1 Welcome screen

The program opens here. Players choose "Play as Guest"; the administrator chooses "Log in as Admin".



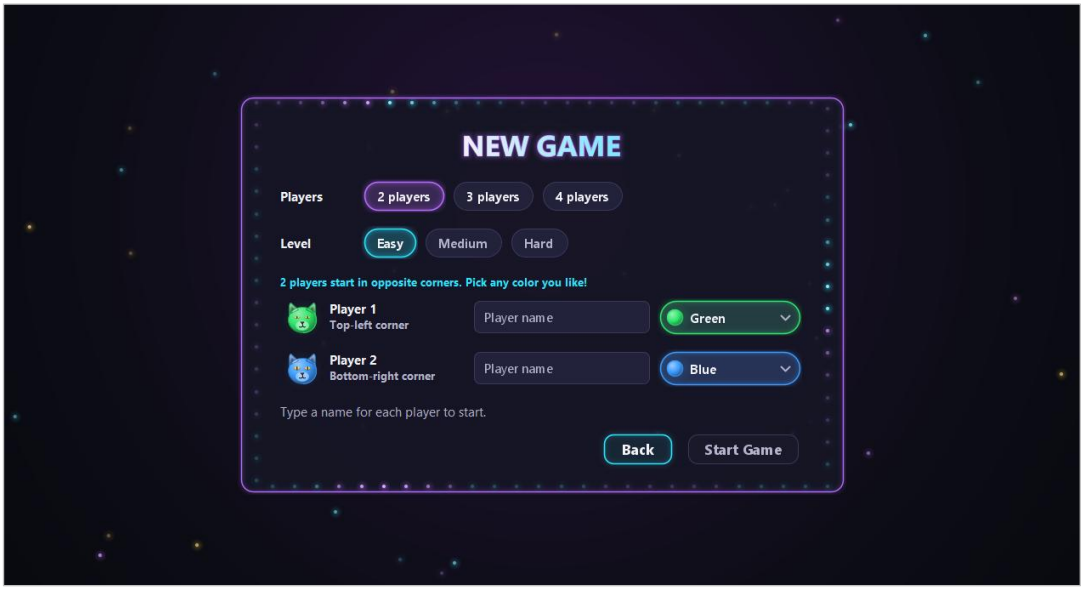
1.2 Main menu (guest)

New Game, Game History, Settings (the gear icon at the top right) and Exit. Question Management is not shown to guests.



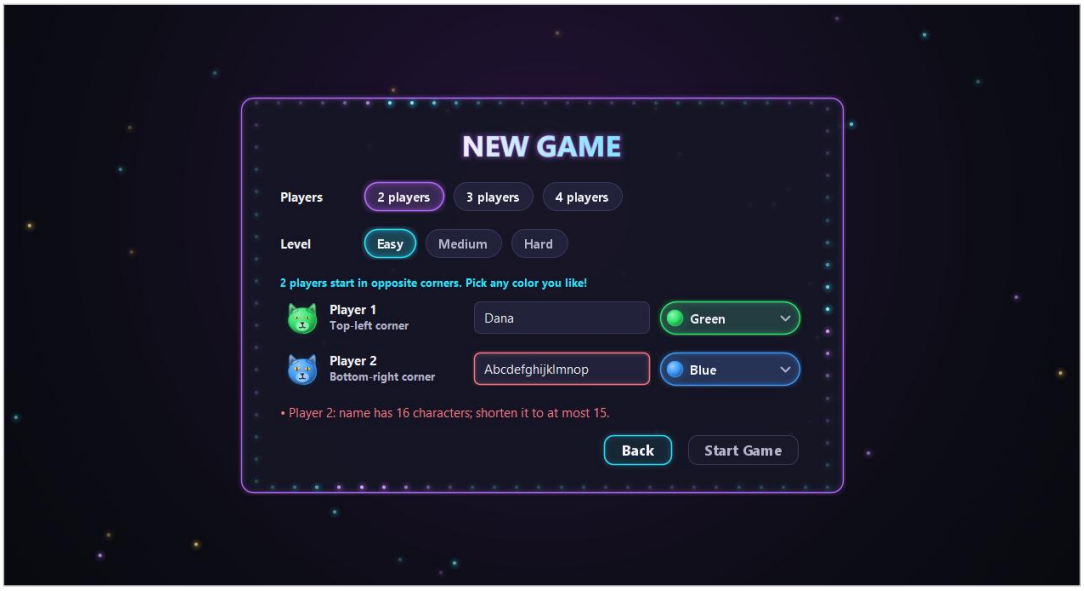
1.3 New game setup

Players choose the number of players and the level. Corners are assigned automatically; each player types a name and picks a color.



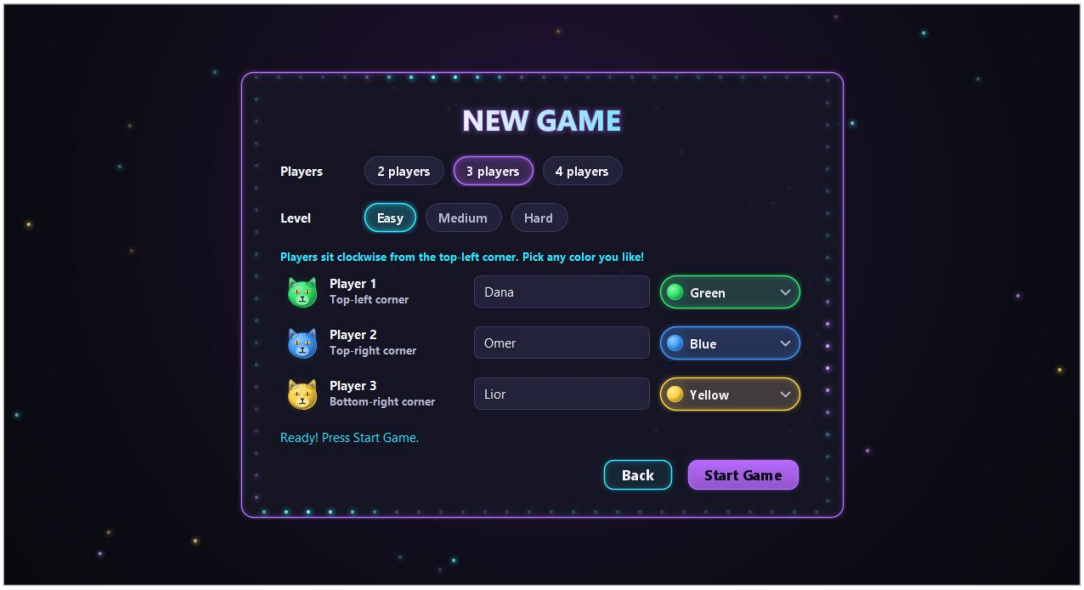
1.4 Setup – a name that is too long

The name field turns red and a message explains the problem and how to fix it. Start Game stays disabled.



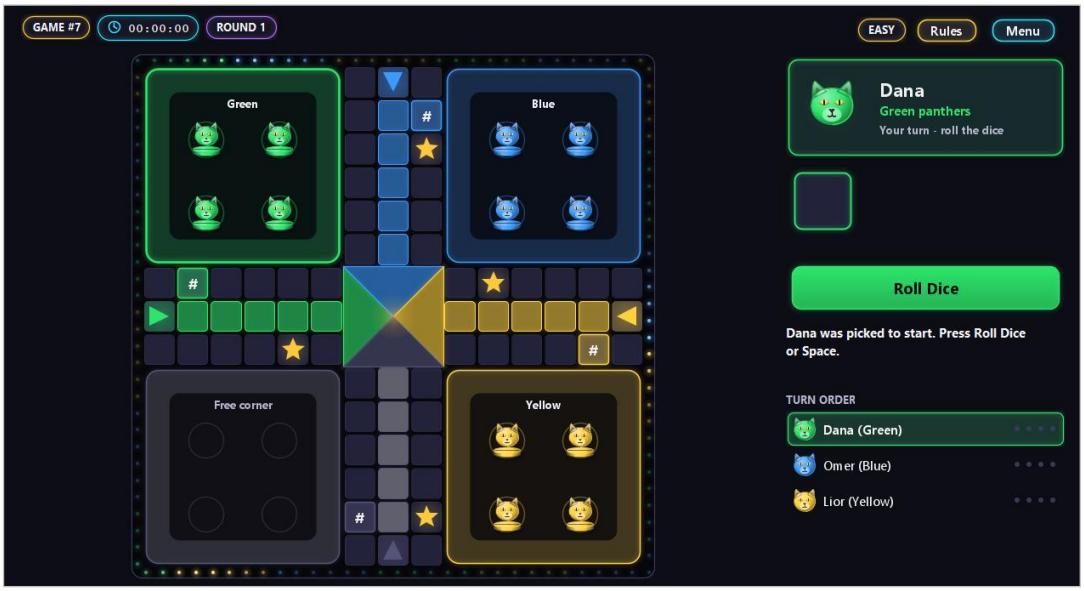
1.5 Setup – ready to start

Three players with valid names and different colors. Start Game is enabled.



1.6 Game screen at the start

The top bar shows the game ID, the game time and the round. The side panel shows whose turn it is, the die, the Roll Dice button and the turn order.



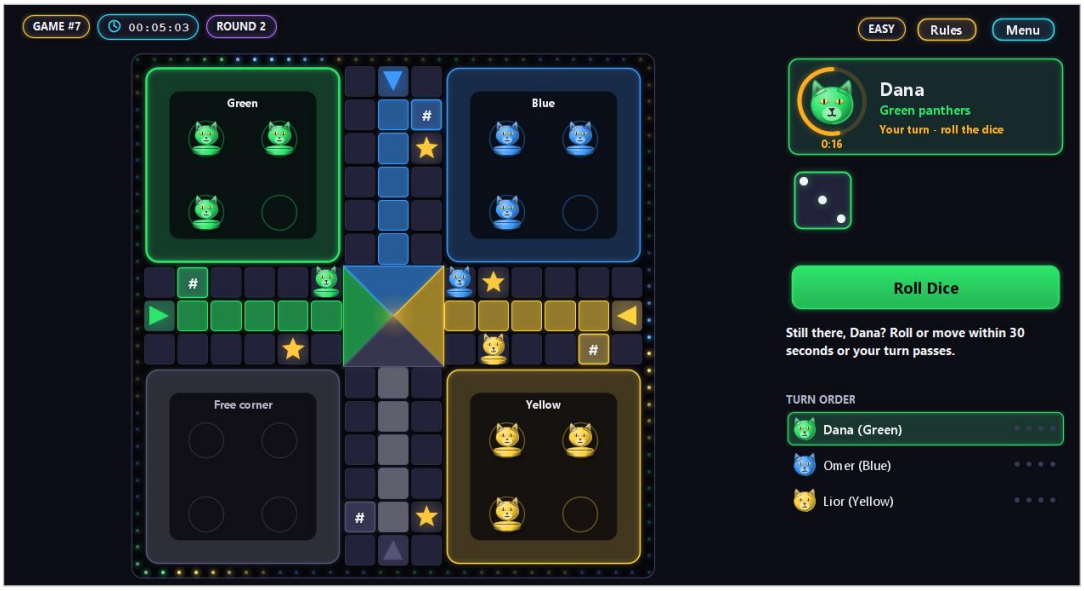
1.7 Choosing a piece after a roll

After rolling a 6, the player's pieces (panthers) that can move glow. The player clicks one, and a message on the side panel explains the move.



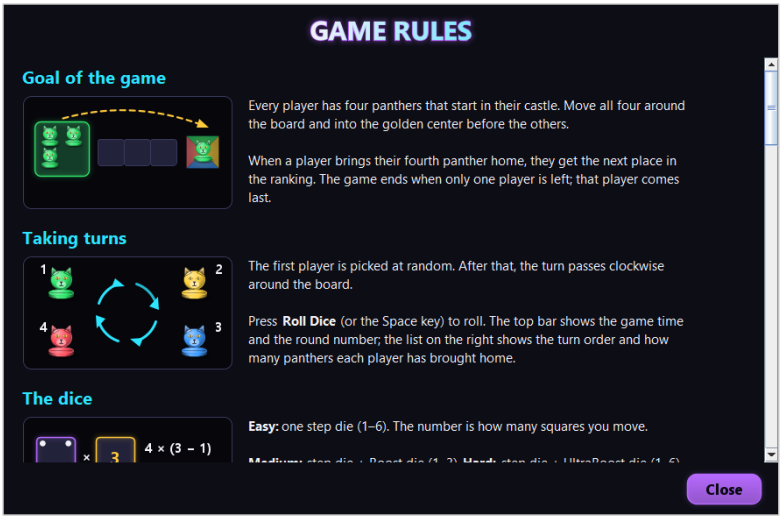
1.8 Inactivity countdown

After 30 seconds without a roll or move, a countdown ring appears around the current player's picture and a reminder is shown. After 60 seconds the turn passes.



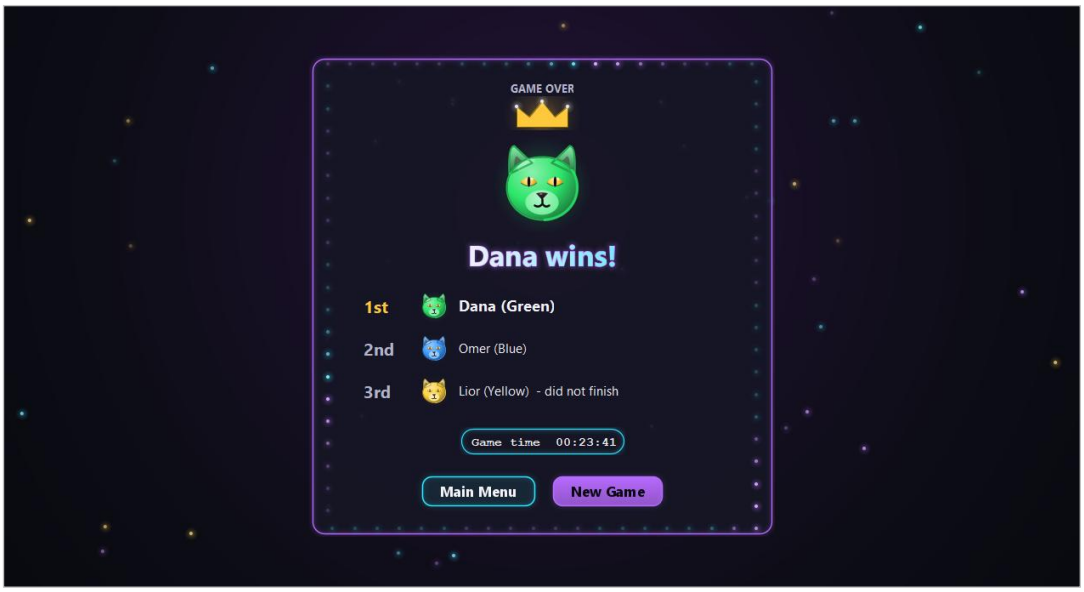
1.9 Rules window

Opened with the Rules button or F1. Each rule has a short text and a picture; the game timer pauses while it is open.



1.10 Game over

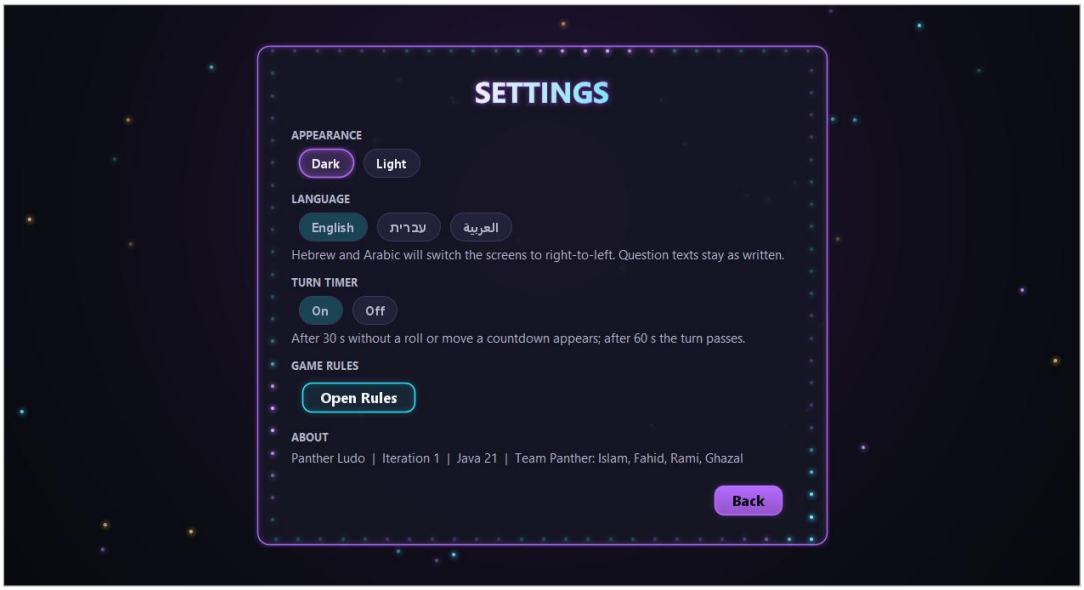
Shows the final ranking and the total game time. Main Menu and New Game lead back to the start of the process.



2. Process 2 – Settings

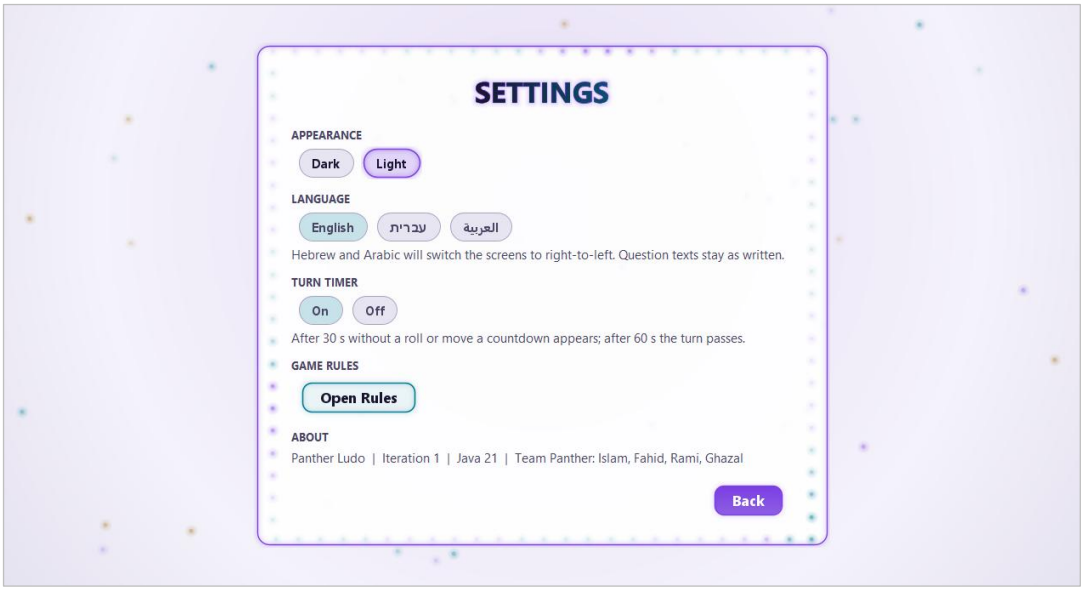
2.1 Settings

Opened with the gear icon at the top right of the main menu: appearance (dark or light), language, turn timer, the rules and information about the game.



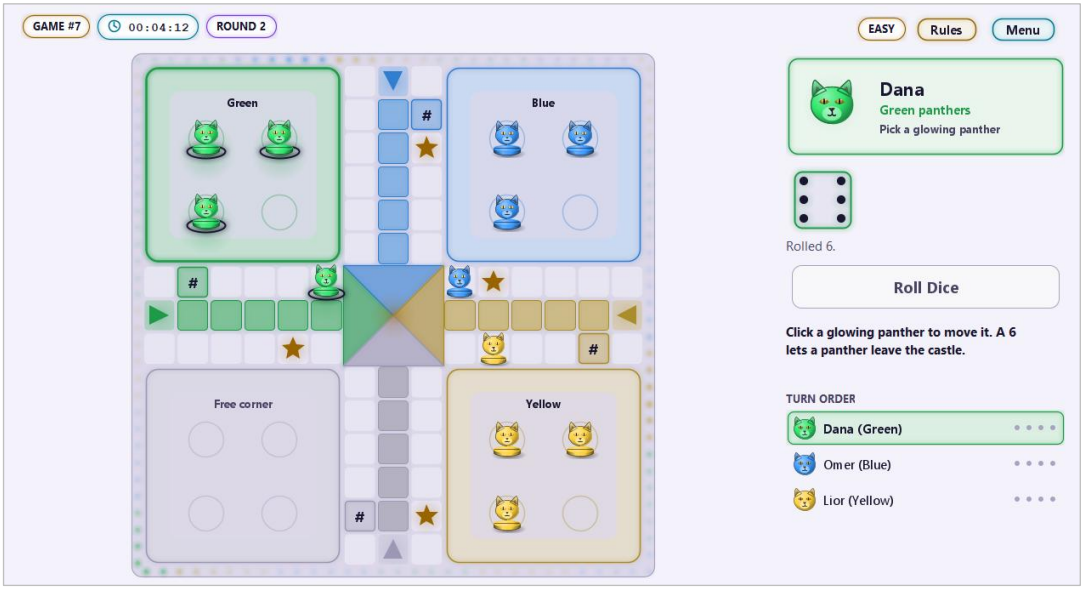
2.2 Settings in light mode

Choosing "Light" switches every screen to the light colors.



2.3 Game screen in light mode

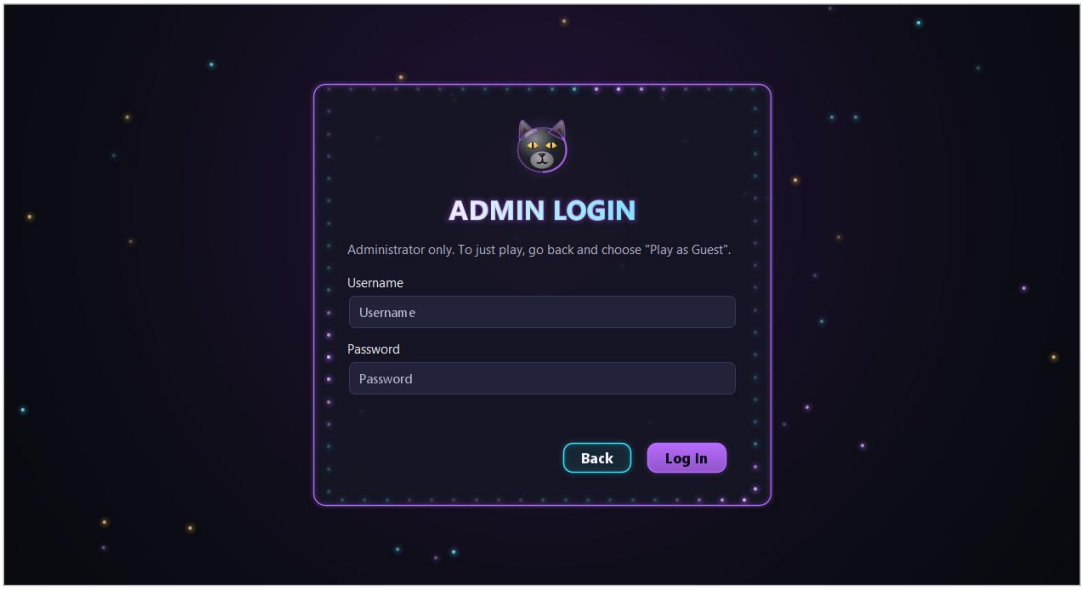
The same game shown with the light colors.



3. Process 3 – Administrator

3.1 Administrator login

Opened from "Log in as Admin" on the welcome screen.



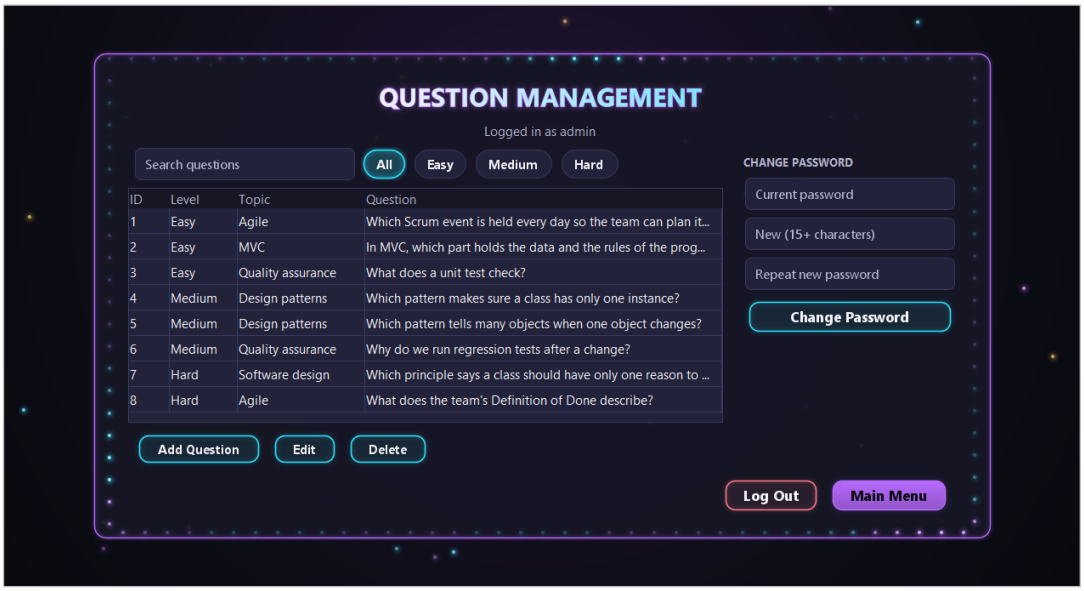
3.2 Main menu (administrator)

After logging in, the menu also shows Question Management and the name of the administrator.



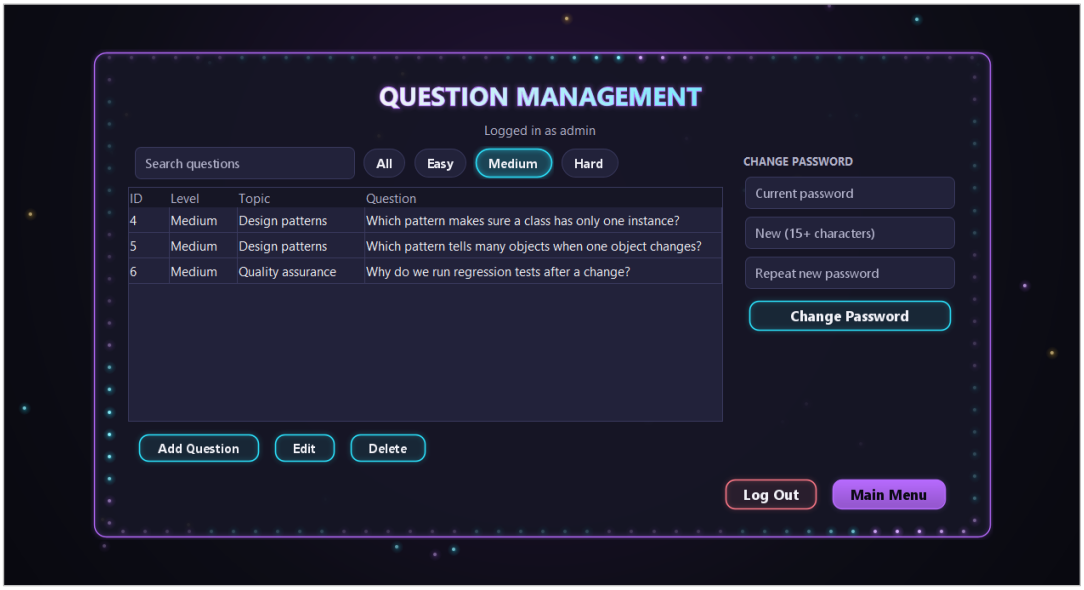
3.3 Question management

The question list with ID, level, topic and text, a search field, level filters, the add, edit and delete buttons and the Change Password form.



3.4 Filtering the questions

Choosing "Medium" shows only the medium questions. Log Out returns to the welcome screen (section 1.1).



4. Tools We Used

Tool	What we used it for and why
Java 21 and Java Swing	The language and screen library set by our SRS; Swing is part of standard Java, so players install nothing extra.
Maven	Builds the runnable JAR file the same way on every computer.
JUnit 5	Unit tests of the game rules, so that changes do not break them.
Visual Studio Code with the Java and Claude extensions	Writing, running and debugging the code.
Git and GitHub	Version control and our shared public repository, where the prototype's JAR file is published.
Claude (Anthropic, Opus 5.5) in Claude Code	AI assistant used as a writing and development aid for the code. Every part was reviewed and run by the team.
Microsoft Word	Writing this document.